DO NOW · AO1 REVISION

Nuclear equations

Atomic structure
Paper 1 · Combined and Separate
7 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 3 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	**Describe how the mass number and atomic number change when an alpha particle is emitted.**	Mass number decreases by 4. Atomic number decreases by 2.	☐
2 ★	**Describe how the mass number and atomic number change when a beta particle is emitted.**	Mass number stays the same. Atomic number increases by 1.	☐
3	**Explain why gamma emission does not change the mass number or atomic number.**	A gamma ray has no mass and no charge.	☐
4	**State the symbol for an alpha particle in a nuclear equation.**	$^{4}_{2}\mathrm{He}$	☐
5	**State the symbol for a beta particle in a nuclear equation.**	$^{0}_{-1}\mathrm{e}$	☐
6 ★	**State the two quantities that must balance in a nuclear equation.**	The total mass number and the total atomic number	☐
7	**State what happens to the element when a nucleus emits an alpha or beta particle.**	It becomes a different element, because the number of protons changes.	☐

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